NATURAL IMAGE STATISTICS FOR SCENE CLASSIFICATION

Akash Chauhan (Roll Number: 23035010653)

Internship Report
Trimester 7

B.Sc. (Honours) Data Science and Artificial Intelligence
Indian Institute of Technology Guwahati, India
akash.chauhan@op.iitg.ac.in

ABSTRACT

This study investigates the statistical properties of natural images and their applicability to image classification tasks. Understanding these properties is important for interpreting how machine learning models perceive and distinguish real-world scenes, with implications for computer vision and image processing applications. Two datasets were analyzed: a Botswana dataset representing open landscapes and an Indian forest dataset representing dense vegetation. Statistical characteristics such as pixel intensity distribution, spatial correlation, frequency content, and edge density were examined to identify distinguishing patterns. Based on these analyses, a set of simple handcrafted features - brightness, standard deviation, edge density, and correlation was used to develop classification models. The binary classifier separated the two datasets with 100% accuracy, but a multi-class extension on the Intel image dataset achieved only 52.5%, due to similarities among certain categories.

Overall, the results suggest that simple statistical features are effective for coarse-level image separation but insufficient for fine-grained classification problems.

Index Terms— Natural images, image statistics, spatial correlation, edge density, scene classification

1. INTRODUCTION

Natural images contain rich statistical structure that reflects the organization of the real world. Unlike random noise, they exhibit strong spatial dependencies between neighbouring pixels, characteristic frequency distributions, and structured edge information. Understanding these regularities is essential for developing robust image analysis and classification techniques in computer vision and machine learning.

In this work, the statistical behaviour of images from contrasting environments is investigated using real-world datasets representing open landscapes and dense vegetation. Properties such as pixel intensity distribution, spatial correlation, frequency content, and edge density are analyzed to identify distinguishing characteristics. Building on these insights, simple handcrafted features are used for two classification tasks: a binary problem between two contrasting environments, and a multi-class problem across six scene categories.

The remainder of this report is organized as follows. Section 2 presents the problem statement and objectives. Section 3 describes the datasets. Section 4 details the methodology and feature extraction. Results are presented in Section 5, followed by discussion in Section 6. Conclusions and future directions are provided in Section 7.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1. Problem Statement

Natural images exhibit structured statistical characteristics arising from the physical organization of real-world environments. These characteristics—spatial dependencies, intensity distributions, and frequency patterns—influence how machine learning and computer vision systems interpret visual information. However, the extent to which such properties vary across different environmental settings, and how effectively they can be exploited for classification, remains an important area of study.

The primary objective is to analyze the statistical properties of natural images and investigate whether these properties differ systematically across visually distinct environments. In particular, the study examines whether simple handcrafted statistical features are sufficient to discriminate between image categories without relying on complex deep learning models.

Two classification tasks were considered:

- **Binary classification** between Botswana images (open landscapes) and Indian Forest images (dense vegetation).
- **Multi-class classification** on the Intel scene dataset across six categories: glacier, sea, street, mountain, buildings, and forest.

The study aims to understand both the strengths and limitations of using low-level statistical image features for scene classification.

2.2. Objectives

The key objectives of this work are summarized as follows:

1. To investigate the statistical properties of natural images, including pixel intensity distributions, spatial correlations, frequency-domain characteristics, and edge density.
2. To compare and analyze statistical differences between environmentally distinct datasets representing open and densely vegetated scenes.
3. To extract meaningful handcrafted statistical features and utilize them for image classification tasks.
4. To develop and evaluate binary and multi-class classification models based on these features.
5. To analyze the performance, limitations, and failure cases of simple statistical-feature-based classification approaches.

3. DATASETS

3.1. Botswana Dataset

The Botswana dataset consists of 50 natural scene images representing open savanna environments — expansive grasslands, open skies, and sparse vegetation. These images typically exhibit large homogeneous regions, smooth intensity transitions, and low texture complexity. A representative selection is shown in Figure 1.

Fig. 1: Sample images from the Botswana dataset.

3.2. Indian Forest Dataset

The Indian Forest dataset contains 50 images of dense tropical forests, jungle canopies, and heavily vegetated terrain. In contrast to the Botswana dataset, these images exhibit high structural complexity, overlapping foliage, varied textures, and strong light-shadow interactions, making it an effective counterpart for comparative analysis. A representative selection is shown in Figure 2.

Fig. 2: Sample images from the Forest dataset.

3.3. Intel Image Classification Dataset

The Intel Image Classification dataset [1], available on Kaggle, was incorporated on the recommendation of the project mentor. It contains approximately 25,000 labeled images across six categories: *glacier*, *sea*, *street*, *mountain*, *buildings*, and *forest*. Categories show substantial visual overlap and intra-class variability, making multi-class classification a useful benchmark for the discriminative capability of simple features. For experiments, 100 images per category (600 total) were used for classification and 50 per category for statistical analysis.

4. METHODOLOGY

All experiments were implemented in Python using Google Colab, with `OpenCV` for image processing, `NumPy` for numerical computation, `Matplotlib` for visualization, and `scikit-learn` for classification and evaluation.

4.1. Statistical Analysis Pipeline

To study the statistical behaviour of natural images, each dataset was processed through the following pipeline:

1. **Image loading and grayscale conversion:** Images were decoded into RGB using OpenCV and converted to grayscale, since most statistical measures depend on intensity.
2. **Histogram analysis:** Intensity histograms were computed for grayscale images and for the individual RGB channels.
3. **Spatial correlation:** Pearson correlation coefficients were computed between pixels at varying lag distances to measure local dependency.
4. **Frequency-domain analysis:** A 2D FFT was applied to each image to inspect low- and high-frequency content.
5. **Edge detection:** The Canny edge detector (thresholds 100, 200) was used; edge density was taken as the fraction of pixels identified as edges.
6. **Aggregation:** Per-image statistics were averaged within each dataset or category for comparative analysis.

4.2. Feature Extraction

Each image was represented by a four-dimensional feature vector summarizing its global statistical and structural properties (Table 1).

Table 1: Handcrafted features used for classification.

Feature	Description
Brightness	Mean grayscale intensity representing overall image luminance.
Standard Deviation	Measure of intensity variation and image contrast.
Edge Density	Fraction of pixels detected as edges using the Canny operator.
Spatial Correlation	Pearson correlation at lag distance 1, capturing local pixel dependency.

Edge density ρ was defined as:

$$\rho = \frac{|\{(i,j) : E(i,j) > 0\}|}{H \times W}, \quad (1)$$

where $E(i,j)$ is the binary edge map and H, W are the image height and width. The features were intentionally simple and computationally lightweight, allowing an investigation into how far low-level statistics alone can support scene classification.

4.3. Classification Models

For the binary task (Botswana vs. Indian Forest), Logistic Regression was used because it is effective on small datasets with low-dimensional features and yields interpretable decision boundaries. For the multi-class task on the Intel scene dataset, multinomial Logistic Regression with softmax was applied first; Support Vector Machine (SVM) and Random Forest classifiers were also evaluated to test whether richer decision boundaries improved performance. All features were standardized to zero mean and unit variance prior to training.

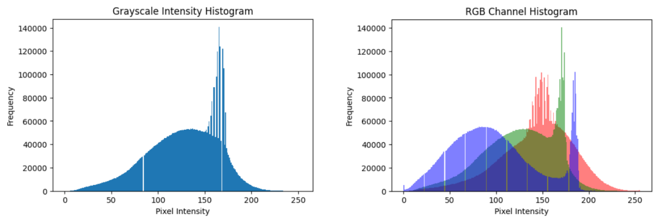
5. RESULTS

5.1. Experimental Setup

For the binary task, 50 images each from the Botswana and Indian Forest datasets were combined (100 total) with an 80:20 stratified split (`random_state=42`), yielding 80 training and 20 testing images. For the multi-class task, 100 images from each of the six Intel categories (600 total) were used; stratified splitting produced 480 training and 120 testing images. The same preprocessing and feature-extraction pipeline was applied to all datasets, and classifier performance was evaluated using accuracy, precision, recall, F1-score, and confusion matrices.

5.2. Pixel Intensity Distribution

Figure 3 shows grayscale and RGB intensity distributions for representative Botswana images. Pixel distributions are highly non-uniform and concentrated in the mid-to-high range, consistent with bright savanna scenes; the blue channel shows a comparatively lower peak.

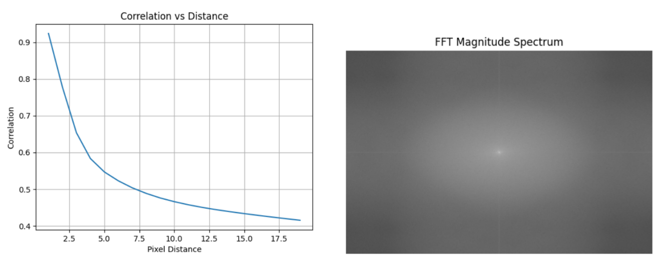


(a) Grayscale histogram (b) RGB channel histograms

Fig. 3: Pixel intensity distributions for representative Botswana images: (a) grayscale histogram and (b) RGB channel histograms.

5.3. Spatial Correlation and Frequency Analysis

Figure 4(a) shows the decay of spatial correlation with increasing pixel distance; Botswana images exhibit strong local dependency (average horizontal correlation $\approx$ 0.967 for neighbouring pixels). The Fourier spectrum in Figure 4(b) shows a bright central region, with image energy concentrated in low-frequency components.

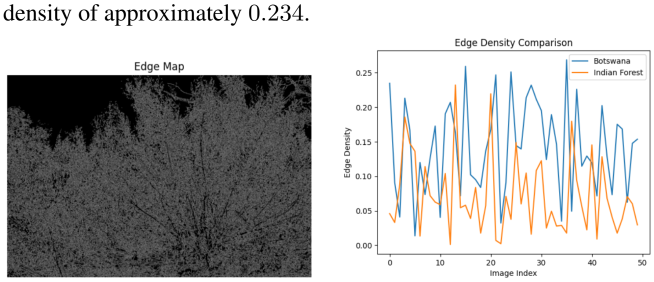


(a) Correlation decay (b) FFT magnitude spectrum

Fig. 4: (a) Spatial Correlation vs Distance and (b) FFT magnitude spectrum.

5.4. Edge Density and Dataset Comparison

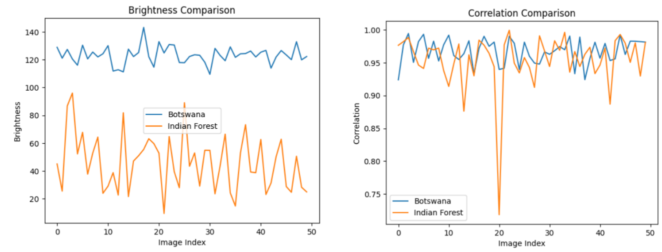
Figure 5(a) presents a representative Canny edge map from the Botswana dataset, while Figure 5(b) compares edge density across the Botswana and Indian Forest datasets. Indian Forest images consistently exhibit higher edge density than Botswana images, while Botswana images contain smoother regions with fewer abrupt intensity transitions. A sample Botswana image produced an edge density of approximately 0.234.



(a) Canny Edge Map (b) Edge Density Comparison

Fig. 5: (a) Canny Edge Map and (b) Edge Density Comparison.

Figure 6 compares average brightness and spatial correlation across the two datasets. Botswana images are generally brighter, while Indian Forest images are darker. Both datasets show high spatial correlation, with Botswana slightly stronger.



(a) Brightness comparison (b) Correlation comparison

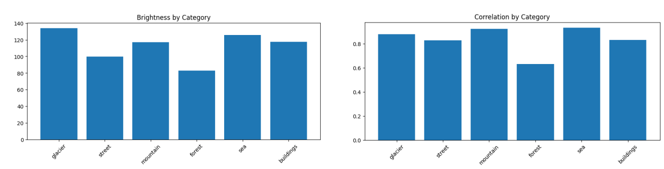
Fig. 6: Comparison of Botswana and Indian Forest datasets

5.5. Category-wise Statistical Analysis

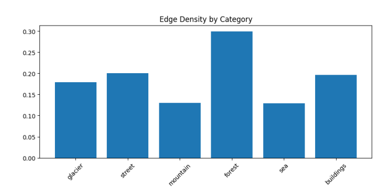
Category-wise statistics for the Intel dataset are summarized in Table 2. Forest images exhibit the lowest brightness and the highest edge density. Glacier and sea categories show higher brightness and lower edge density, while buildings and street fall in an intermediate range. Figure 7 visualizes these variations and highlights the statistical overlap between several scene classes.

Table 2: Average image statistics for Intel scene categories.

Category	Brightness	Correlation	Edge Density
Glacier	133.32	0.886	0.165
Sea	126.15	0.929	0.149
Street	98.81	0.822	0.205
Mountain	122.73	0.915	0.142
Buildings	115.15	0.794	0.217
Forest	84.78	0.629	0.311



(a) Brightness (b) Correlation



(c) Edge density

Fig. 7: Per-category statistical comparisons across the Intel dataset.

5.6. Binary Classification: Botswana vs. Indian Forest

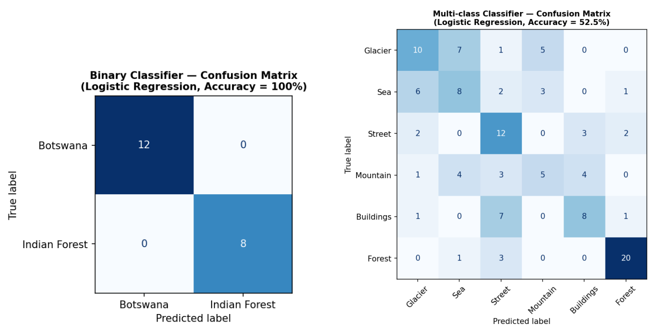
The Logistic Regression classifier achieved **100% accuracy** on the binary classification task, correctly classifying all Botswana and Indian Forest test samples without any misclassifications. The corresponding confusion matrix is shown in Figure 8(a). Both precision and recall were equal to 1.0 for each class.

5.7. Multi-class Classification: Intel Scene Categories

The multi-class Logistic Regression classifier achieved an overall accuracy of **52.5%** on the Intel dataset (Table 3). Among all categories, forest images achieved the highest F1-score of 0.83, while mountain images obtained the lowest at 0.33. The confusion matrix in Figure 8(b) shows frequent misclassification among glacier, sea, and mountain, and additional confusion between the buildings and street categories.

Table 3: Multi-class Classification Performance using Logistic Regression.

Category	Precision	Recall	F1-Score	Support
Glacier	0.50	0.43	0.47	23
Sea	0.40	0.40	0.40	20
Street	0.43	0.63	0.51	19
Mountain	0.38	0.29	0.33	17
Buildings	0.53	0.47	0.50	17
Forest	0.83	0.83	0.83	24
Accuracy		**52.5%**		
Macro Avg	0.51	0.51	0.51	120



(a) Binary Classification (b) Multi-Class Classification

Fig. 8: Confusion Matrices

6. DISCUSSION

The results confirm that natural images exhibit consistent statistical structure: high spatial correlation between neighbouring pixels, and energy concentrated in low-frequency components of the Fourier spectrum—both consistent with the natural image statistics literature.

The binary classifier achieved 100% test accuracy because the two datasets differ substantially: Botswana images show smooth intensity transitions, higher brightness, and lower complexity, while Indian Forest images contain dense textures and significantly higher edge density. The classifier relied primarily on brightness and edge density, which were the most discriminative features between the datasets. This demonstrates that simple statistical features are highly effective when scene categories exhibit strong visual and structural differences. However, with only 20 test samples this accuracy carries inherent statistical uncertainty—a different random split could shift the result—and cross-validation on the full dataset would give a more reliable estimate of generalization.

The multi-class task proved considerably more challenging at 52.5% accuracy. The confusion matrix shows substantial overlap among glacier, sea, and mountain categories, which share similar statistics (Table 2). Forest was the easiest to classify (F1 = 0.83) because of its distinctly low brightness and high edge density, whereas mountain (F1 = 0.33) suffered from overlap with glacier and sea. Buildings and street were also frequently confused, as both contain strong geometric edges. SVM and Random Forest produced only marginal improvements, suggesting the bottleneck lies in the representational capacity of the features rather than the choice of classifier.

7. CONCLUSION AND FUTURE WORK

The main takeaway is that the discriminative power of low-level statistical features depends strongly on how distinct the underlying scene categories are. When environments differ substantially in brightness, texture, and structural complexity, a four-dimensional handcrafted vector achieves perfect separation; when categories share global statistics, the same features collapse and no choice of classifier (Logistic Regression, SVM, or Random Forest) recovers the loss. The bottleneck observed here is therefore representational rather than algorithmic.

A useful next step is to enrich the feature representation: texture descriptors (e.g., GLCM, LBP), local rather than global statistics, and multi-scale frequency analysis would all capture structure that the current global descriptors discard. Convolutional Neural Networks offer a learned alternative capable of capturing hierarchical scene semantics. Larger datasets and k-fold cross-validation would also tighten the reliability of the estimates.

8. ARTIFACTS AND DEMONSTRATIONS

- **Code Repository:** GitHub repository containing all analysis, feature extraction, and classification code: `https://github.com/akashhh-codes/natural-image-statistics`
- **Datasets:** Botswana and Indian Forest datasets used in this project, along with the publicly available Intel Image Classification dataset: `https://www.kaggle.com/datasets/puneet6060/intel-image-classification`

Note: AI-assisted tools were used for writing and formatting. All analysis, code, and interpretation are the author's own work.

Acknowledgment: This work was carried out with the mentorship of Neeraj Kumar Sharma, Principal Investigator, SPIN Lab, IIT Guwahati.

9. REFERENCES

[1] Puneet Bansal, "Intel image classification," Kaggle Dataset, 2019, `https://www.kaggle.com/datasets/puneet6060/intel-image-classification`.

[2] E. P. Simoncelli and B. A. Olshausen, "Natural image statistics and neural representation," *Annual Review of Neuroscience*, vol. 24, pp. 1193–1216, 2001.

[3] D. J. Field, "Relations between the statistics of natural images and the response properties of cortical cells," *Journal of the Optical Society of America A*, vol. 4, pp. 2379–2394, 1987.

[4] A. Torralba and A. Oliva, "Statistics of natural image categories," *Network: Computation in Neural Systems*, vol. 14, pp. 391–412, 2003.

[5] G. J. Burton and I. R. Moorhead, "Color and spatial structure in natural scenes," *Applied Optics*, vol. 26, pp. 157–170, 1987.

[6] A. J. Bell and T. J. Sejnowski, "The independent components of natural scenes are edge filters," *Vision Research*, vol. 37, pp. 3327–3338, 1997.

[7] W. S. Geisler, J. S. Perry, B. J. Super, and D. P. Gallogly, "Edge co-occurrence in natural images predicts contour grouping performance," *Vision Research*, vol. 41, pp. 711–724, 2001.

[8] D. Kersten, "Predictability and redundancy of natural images," *Journal of the Optical Society of America A*, vol. 4, pp. 2395–2400, 1987.

[9] D. Ruderman and W. Bialek, "Statistics of natural images: Scaling in the woods," *Physical Review Letters*, vol. 73, pp. 814–817, 1994.

[10] D. L. Ruderman, "Origins of scaling in natural images," *Vision Research*, vol. 37, pp. 3385–3398, 1997.

[11] D. A. Forsyth and J. Ponce, *Computer Vision: A Modern Approach*, Prentice Hall, New Jersey, USA, 2002.